

# Daniel He

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## Education

**Western University** - BSc., Honours Specialization in Computer Science *September 2024 - May 2028 Expected*  
With Admission Scholarship, Scholarship of Distinction (\$2500) | **GPA: 3.50/4.00**

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## Work Experience

**OpsGuru (AWS Premier Partner)** - Software Engineering Intern *May 2026 - August 2026*  
*AWS AI/ML Infrastructure & Agent Systems*

- Built TypeScript/React.js frontend features for 10 customer-facing AI agents, deployed via **Terraform** and AWS
- Designed and engineered a customer-facing AI demo agent to query customer policies using **AWS Bedrock Agentcore**, Python, and **LangGraph**. Implemented an error recovery loop that salvaged ~90% of failed queries

**Western University** - Undergraduate Research Assistant *October 2025 - May 2026*  
*LLM Hallucination - Dr. Apurva Narayan*

- Co-authored *HalluciBench*, an LLM hallucination benchmarking paper, submitted to COLM 2026
- Designed and implemented an LLM-as-a-judge benchmarking pipeline in Python with **Ollama** to evaluate hallucination rates across **29 LLMs**, incorporating Hyperparameter Tuning and **Prompt Engineering**
- Implemented a **Reinforcement Learning**-inspired **Synthetic Data** generation pipeline from a research paper to create a **1000-sample** dataset for evaluating LLM refusal calibration and hallucination rates

**Scotiabank** - Software Engineering Intern *May 2025 - August 2025*  
*IT&S Capital Markets/Data Analytics*

- Refactored a broken **Airflow** DAG pipeline to store vector embeddings in **ChromaDB** with persistence to **Google Cloud Storage**, migrating embedding logic into a Cloud Function. Cut p95 latency by ~90% and resolved a 6-month production issue, enabling a scalable RAG knowledge base for the bank's internal knowledge bot
- Enhanced internal research summarization pipeline by extending PDF parsing support from 2 to 4 document formats, automated deployment using **CI/CD** build configuration tools, and documented using Jira and Confluence
- Built a **Python** cyclomatic complexity analyzer using SonarQube and Selenium, reducing analysis time by ~85%

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## Projects

**Dex2** ([GitHub Link](#), [Video-Demo](#)) *February 2026*

- Built a **TypeScript/React.js** Chrome extension that auto-captures page content to trigger context-aware agent actions (edit slides, open tabs, draft emails) using **MongoDB** for embeddings and Supabase (**PostgreSQL**)
- Implemented AI **Agent Orchestration** using FastAPI for hybrid retrieval and **Tool Calling** with background workers and concurrency to keep chat non-blocking. Used **LangChain** for chunking and retrieval
- Built a Google Slides agent combining **OAuth** to generate and edit slides directly in live presentations

**Mini-v** ([GitHub Link](#)) *April 2026*

- Built a **C++** LLM inference server with micro-batching and continuous batching, achieving **2.5x throughput** improvement (6.5 to 16.8 req/s) across concurrent load with 100% request success rate

**Say Less** ([Devpost Link](#), [Live-Demo](#)) *November 2025*

*Hack Trent 2025 - MLH Best Use of ElevenLabs (21 total submissions)*

- Created a website that converts American Sign Language (ASL) to speech and speech to text in real time
- Spearheaded development of a **FastAPI** backend with **RESTful API** endpoints, integrating a fine tuned **MediaPipe** model (~80% accuracy on ASL to text). Containerized with **Docker** and deployed on **Render**

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## Skills

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL | Spoken Languages: English, Chinese (Mandarin)  
Frameworks/Libraries: Next.js, React.js, FastAPI, MediaPipe, Tensorflow, PyTorch, SQLAlchemy, LangChain, LangGraph  
Developer Tools: Docker, Git, Vercel, GCP, Apache Airflow, AWS, Ollama, Linux, Claude Code, Cursor, Terraform  
Databases: PostgreSQL (Supabase), SQLite, MongoDB, ChromaDB